Fiscal Year:	FY 2011	**Task Last Updated:**	FY 03/20/2011
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	05/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Expedition 19/20 NOTE: extension to 5/31/2013 is in process, per PI (Ed., 3/6/12) NOTE: End date is now 5/31/2012 (from 5/31/2010) per NSSC information and F. Szofran/MSFC (10/2009) NOTE: Changed to FLIGHT project, per F. Szofran/MSFC (10/2009) NOTE: End date changed to 5/31/2010 (from 5/31/2009) per PI (12/08)		
Key Personnel Changes/Previous PI:			
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Performance Goal Text:	
Task Description:	Supercooling, the ability of a liquid held at temperatures below its melting temperature without crystallizing, was first discovered by Fahrenheit in 1724 when he was studying the freezing of water. It wasn't until 1952 that supercooling in liquid metals was demonstrated by David Turnbull. That liquids can be supercooled indicates that there is a barrier to the crystallization of the liquid; this is called the nucleation barrier. Nucleation is the first step in most first-order phase transitions, like the freezing of water. Nucleation is important in processes ranging from the crystallization of liquids to the development of bubbles in carbonated beverages to biological processes. It is even believed to be central in phase transitions that occurred in the early universe. An ability to control nucleation would be valuable for the development of new materials for advanced applications. In almost all cases nucleation data are described by the classical theory of nucleation, first developed in the late 19th century. It is now recognized however that there are serious problems with aspects of that theory in some regimes of application. Also, it does not properly describe nucleation when the chemical concentrations of the initial and final phases are different. Finally, recent evidence indicates that nucleation is often coupled to other processes and phase transitions, something that cannot be described readily within the classical theory. The task of the continuing ground-based research on this contract is the investigation of a coupling between structural ordering in the liquid and associated thermophysical properties, and with the nucleation barrier, and with other phase transitions that are being discovered that occur in supercooled liquids. Studies are focused on transition metals and their alloys. Ground-based measurements of the structure, specific heat, density, viscosity, surface tension, and emissivity in equilibrium and supercooled liquids are made using electrostatic levitation containerless processing techniques. In cooperation with ESA these data will be combined with studies made on the International Space Station to develop a better understanding of the complicated nucleation processes, and other phase transitions, that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The WU-BESL (Washington University Beamline Electrostatic Levitation) technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions. The discovery of possible new phase transitions in the liquid will lead to a better understanding of glass formation.
Task Progress:	Last year we designed and constructed a new Electro-Static Levitation (ESL) facility for synchrotron scattering studies of liquids. The WU-BESL (Washington University Beamline ESL) was designed to be portable, operated by one person, and optimized for X-ray diffraction and thermophysical property measurements of equilibrium and supercooled liquids. This year, we have improved the WU-BESL, constructing a compact alignment table that will allow the facility to be used on a wide range of synchrotron beamlines, and constructing a sample carousel that minimizes the amount of time that the vacuum chamber must be brought up to air to change samples. We have refined our processing technique, using a tungsten post to pre-process samples before beginning levitation processing, which leads to much better sample stability during levitation. We have also improved the sample stability during levitation. Finally, we have improved our density measurement technique, allowing precise investigations of changes in liquid density with temperature and small changes in alloy composition. In August, 2010, 73 different alloy systems were processed at the Advanced Photon Source. Those data are now being analyzed. A brief description of key results obtained over the past year follows: 1. One of the samples planned for experiments on the International Space Station by members of the ESA team to which my project is linked is Si50Ge50. Theoretical and computer studies have shown that silicon (Si) undergoes a novel liquid/liquid phase transition below the melting temperature, if crystallization can be avoided. Earlier experimental scattering studies made by us using the Marshall ESL gave no evidence for this transition down the supercooling limit. It was thought that it might be easier to observe it in an alloy of Si and germanium (Ge), since Ge has a much lower melting temperature. Alloying them would lower the melting temperature (or liquidus temperature for alloys) and allow lower temperatures to be explored without crystallizing. In 2010, we obtained measurements of the Si50Ge50 liquid down to 1226K, 150oC lower than was possible with the pure Si liquid. Interestingly, this alloy shows much less medium-range order than the Si does. Further, this order shows no measurable change with decreasing temperature. This raises concerns as to whether the liquid/liquid phase transition can be observed in this alloy. Technical difficulties made it impossible to process pure Ge. We are working to design methods that will allow that in future WU-BESL runs at the APS. 2. There is significant interest in finding materials that can be used to run engines at higher temperatures to improve efficiency. Frequently, however, the phase diagrams for these materials are only partially known, or not known at all, due to the considerable difficulties in containing samples at such high temperatures for study. Since processing is carried out without a container in WU-BESL, this provides a new approach to probing these high temperature domains. To demonstrate this, we explored the behavior of liquids of hafnium (Hf) and rhodium (Rh) and hafnium and platinum (Pt) near the eutectic composition (where the liquidus temperature has a local, and possibly global, minimum, depending on the details of the phase diagram). For Hf-Rh, our studies have demonstrated that the eutectic composition is Hf74Rh26, in contrast with previous claims that placed it at Hf79Rh21. We have also found that the number density is a maximum at the new eutectic composition, supporting arguments that the eutectic liquid is more stable and dense than liquids of other compositions. In agreement the number density at the previously claimed eutectic composition is much higher. There is very little known about Hf-Pt liquids and the phase diagram at high temperatures. We have determined the liquidus temperatures for these alloys and obtained the first structural data of the liquids. We have also obtained diffraction data for the primary crystallizing phases. We are currently working to use this data to solve the structures of these phases. When complete, we will have made significant progressing in mapping out the first high temperature phase diagram for this alloy. 3. We have shown that liquids can have extended structural order, demonstrating the presence of a low angle scattering peak (pre-peak) in a zirconium (Zr) - platinum liquid, Zr80Pt20. The scattering data were used to construct an atomic structure for this liquid, which showed an increase in icosahedral short-range ordering with decreasing temperature. The pre-peak was shown to originate primarily from correlations between the Pt atoms, a next-nearest-neighbor effect. This demonstrates the existence of order over a longer length scale than would be expected in a highly mobile liquid. From data collected in 2010, we have demonstrated, however, that this is common when the heat of mixing (a measure of how strongly unlike atoms bond) is negative and large. For example, we have observed pre-peaks in alloys of Zr with iridium (Ir), Zr-Ir, and with gold (Au), Zr-Au, both alloys that have strong negative heats of mixing. However, some alloys satisfy this criterion but show no pre-peaks, such as Zr-Pd (palladium, Pd) and Zr-Rh. Our work builds the case, however, that this does not indicate the lack of medium range order. Instead the pre-peaks cannot be seen due to a lack of X-ray contrast when the two atomic species have similar atomic numbers. It is likely, then, that medium-range order is quite

	common. This is important since it can have a profound impact on the crystal nucleation barrier, limiting supercooling and allowing access to metastable crystal and amorphous phases during cooling. Our ground-based work is laying the foundation for investigations of the structures of liquids and how that relates to phase transitions, supporting the planned ISS experiments, and pioneering a new way to determine high-temperature phase diagrams. These results of these studies, therefore, have significant basic as well as practical interest.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
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